CHAPTER-4

CATALYSIS

Q. What is (i) Catalyst and (ii) Catalysis? Give example. [2016.

(i) Catalyst:

Catalyst is any substance which can change the speed of the reaction without being used up in that reaction.

(ii) Catalysis:

The phenomenon in which catalyst can change the speed of the reaction without being used up in that reaction is known as catalysis.

Example:

$$2KMnO_4 + 3H_2SO_4 + 5 \begin{matrix} COOH \\ | \\ COOH \end{matrix} \longrightarrow K_2SO_4 + 2MnSO_4 + 8H_2O + 10C$$

In this reaction, Mn^{2+} act as a catalyst and this process is called catalysis.

Q. What is auto catalysis? [2015].

Auto catalysis: The phenomenon in which one of the product itself acts as a catalyst is known as auto catalysis.

Example:

$$2KMnO_4 + 3H_2SO_4 + 5\begin{matrix}COOH\\COOH\end{matrix} \longrightarrow K_2SO_4 + 2MnSO_4 + 3H_2O + 10CO_2$$

The speed of reaction increases as the reaction progresses. This is due to the presence of Mn^{2+} ions which get formed in the reaction.

Q. Explain the characteristics of catalysis? [2018] [2016] [2015]

Characteristics of catalysis:

I. The catalyst remains unchanged in mass and chemical composition:

At the end of the reaction, a change in its physical state, color etc may occur but the catalyst remains unchanged in mass and chemical composition.

II. Small amount of catalyst is needed:

Usually a small quantity of the catalyst bring about a large amount of chemical transformation.

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For example, one gram of copper in 10^8 litres, catalysis the oxidation of $NaHSO_3$ by air.

(iii) Effect of the equilibrium of a reversible reaction:

In case of reversible reactions, the catalyst does not influence the composition of reaction mixture at equilibrium.

For example, the decomposition of Hydrogen iodide at equilibrium shows not to change the concentration of reactor or product.

$$2HI \rightleftharpoons H_2 + I_2$$

(iv) Inability to start a reaction:

According to Ostwald, a catalyst can not start a reaction but can only decrease or increase the reaction rate.

(v) Specific Nature of a catalyst:

A particular substance can act as a catalyst only in a particular reaction and not in all.

For example, MnO_2 may catalyse the decomposition of $KClO_3$ but not $KClO_4$.

(vi) Nature of the products is unaltered by the presence of the catalyst:

H_2 and N_2 always combine to give NH_3 in the absence or presence of the catalyst. Some exceptions are-

$$CO + 3H_2 \xrightarrow{Ni} CH_4 + H_2O$$

$$CO + H_2 \xrightarrow{Cu} HCHO$$

$$CO + 2H_2 \xrightarrow{ZnO + Cr_2O_3} CH_3OH$$

(vii) Optimum temperature: There is a particular temperature at which the efficiency of a catalyst is most marked. This temperature is called optimum temperature.

(viii) Action of promoters. The addition of small amount of foreign substances, sometimes increases the activity of the catalyst.

For example, in manufacturing of ammonia by Haber's process Fe act as a catalyst while Mo acts as a promoter.

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(ix) A catalyst is poisoned by certain substances: Some substance inhibited or fully destroyed the catalytic property of a catalyst.

Example: The activity of iron catalyst is destroyed by the presence of H_2S or CO in synthesis of ammonia by Haber's process.

Q: What do you understand by the terms catalytic poisons and catalytic promoters? [2018] [2017]

Catalytic promoters:

A catalytic promoter is a substance that is usually mixed in small quantities with catalysts. Mixing this substance increases the efficiency of the catalyst. In simple words, it helps in improving the overall chemical reaction. However the catalytic promoter has little or no catalytic effect or properties of its own. They are also known as activators.

Example:

In Haber's process for the synthesis of ammonia traces of molybdenum increases the activity of finely divided iron which acts as a catalyst hence act like a catalytic promoter.

$$N_2 + 3H_2 \xrightarrow{Fe/Mo} 2NH_3$$

Catalytic poisons;

Catalytic poisons are those substances which inhibit or destroy the activity of the catalyst by their presence.

Example:

i. The activity of Iron catalyst is destroyed by the presence of H_2S or CO in synthesis of ammonia by Haber's process.

ii. The platinum catalyst used in the oxidation of hydrogen is poisoned by CO.

Q. What are the different types of catalysis? Give examples of each type. [2016] [2015]

There are two types of catalysis:

(a) Homogeneous catalysis:

In homogeneous catalysis, the catalyst and reactants are in the same phase.

(i) Examples from gaseous phase—

$$2SO_2 + O_2 \xrightarrow{NO} 2SO_3.$$

(ii) Example from liquid phase—

$$CH_3COOC_2H_5 + H_2O \xrightarrow{H^+} CH_3COOH + C_2H_5OH.$$

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iii. In solid phase, catalyst and the reactants are not homogeneously occuar.

(b) Heterogeneous catalysis: In heterogeneous catalysis, the catalyst and reactants are in different phase.

Examples are-

(i) Heterogeneous catalysis involving solid reactants:

$$2KClO_3(s) \xrightarrow{MnO_2} 2KCl + 3O_2$$

(ii) Heterogeneous ~~catalysis~~ catalysis involving liquid reactant:

$$2H_2O_2(l) \xrightarrow[Au]{Pt \text{ or }} 2H_2O + O_2$$

(iii) Heterogeneous catalysis involving gaseous phase:

$$2SO_2(g) + O_2(g) \xrightarrow{Pt(s)} 2SO_3(g)$$

Q. Discuss the theories and mechanism of homogeneous catalysis. [2017]

The various postulates for theory of homogeneous catalysis are -

I. The catalyst first formed an intermidiate compound with the reactant.

$$A + X \underset{k_2}{\overset{k_1}{\rightleftharpoons}} AX$$

Where, A is the substrate, X is the catalyst, AX is the intermidiate compound, k_1 and k_2 are the velocity constant for the forward and backward reactions.

II. The intermediate compound then reacts with other reactant molecule to form the product and catalyst.

$$AX + B \xrightarrow{k_3} AB + X$$

This reaction is slow and rate determining step.

Thus,

rate of reaction $= k_3 [AX] [B]$ ——— (1)

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III. The catalyst X which is regenerated in last step may further undergo step (I) and (II) to form more and more of the products.

Thus the rate of homogeneous catalytic reactions depends upon the concentration of catalyst X.

Now, if we assume steady state approximation for [AX] we get –

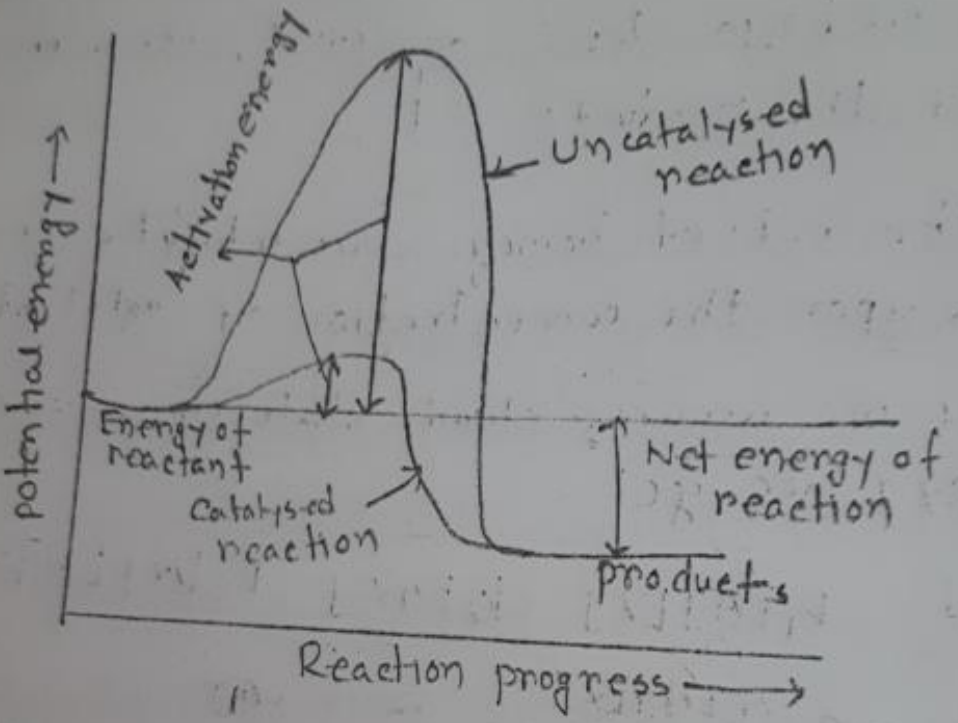
$$\frac{d[AX]}{dt} = k_1[A][X] - k_2[AX] - k_3[AX][B] = 0$$

$$\Rightarrow AX = \frac{k_1[A][X]}{k_2 + k_3[B]} \quad \text{——— (II)}$$

Substituting the values of [AX] in equation (II) we get –

$$\text{Rate of reaction} = \frac{k_1 k_3 [A][X][B]}{k_2 + k_3[B]} \quad \text{——— (III)}$$

According to the theory of homogeneous catalysis, the catalyst can provide an alternative path for a reaction to occur at lower activation energy.



Examples:

$$2SO_2 + O_2 \xrightarrow{NO(g)} SO_3$$

Mechanism:

$$2NO + O_2 \longrightarrow 2NO_2 \text{ (Intermidrate compound)}$$

$$NO_2 + SO_2 \longrightarrow SO_3 + NO$$

Here, in first step the catalyst NO combine with a reactant O_2 and form intermidiate compound $2NO_2$. This intermidiate compound then react with another reactant and form product and free the catalyst.

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Q. Derive the rate equation of homogeneous catalysis and show how the rate of such reaction related to the concentration of catalyst.

According to the theory of homogeneous catalysis-

The catalyst first formed an intermidiate compound with the reactant :

$$A + X \underset{k_2}{\overset{k_1}{\rightleftharpoons}} AX \quad \text{---------- (I)}$$

Where A is the reactant molecule, X is the catalyst and AX is the intermidiate compound. k_1 and k_2 are velocity constants for the forward and reverse reaction.

The intermidiate compound then reacts with other reactant molecule to form product and catalyst -

$$AX + B \xrightarrow{k_3} AB + X \quad \text{---------- (II)}$$

This reaction is slow and rate determining step.

$\therefore$ Rate of the reaction $= k_3 [AX][B]$ ---------- (II)

If we assume steady state approximation for [AX] we get -

$$\frac{d[AX]}{dt} = k_1[A][X] - k_2[AX] - k_3[AX][B] = 0$$

$$\Rightarrow [AX] = \frac{k_1[A][X]}{k_2 + k_3[B]}$$

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Substituting the value of [AX] in equation (iii) we get-

$$\text{Rate of equation} = \frac{k_1 k_3 [A][X][B]}{k_2 + k_3[B]} \quad \text{——— (iv)}$$

Limiting cases:

(i) If $k_2 \ll k_3[B]$ then equation becomes-

$$\text{rate of equation} = \frac{k_1 k_3 [A][X][B]}{k_3[B]}$$

$$= k_1 [A][X] \quad \text{——— (v)}$$

(ii) If $k_2 \gg k_3[B]$ then equation (iv) becomes-

$$\text{rate of reaction} = \frac{k_1 k_3 [A][X][B]}{k_2} \quad \text{——— (vi)}$$

From equation (v) and (vi), it is seen that, the rate of a homogeneous catalytic reaction depends upon the concentration of catalyst and with increasing the concentration of catalyst, increases reaction rate and vice-versa.

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Q. Explain the function of a catalyst in terms of Gibb's free energy of activation. [2018]
or, How does catalyst work - explain with the help of energy diagram. [2017]

A catalyst prod provides alternate path for the reaction that is having a lower free energy of activation. The lowering of free energy is probably attributed to a decrease in energy of activation or a higher frequency factor or both.

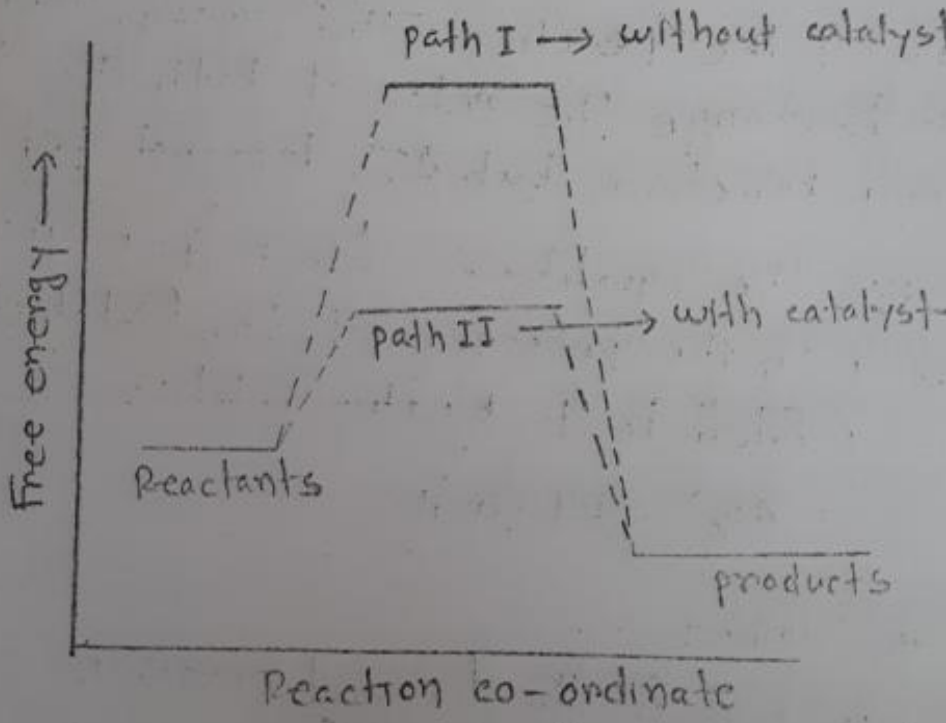


Fig.: Lowering of Gibbs free energy of activation of a reaction by a catalyst.

From the activated complex theory without catalyst we get-

$$k_f = \frac{k_B T}{h} \exp(-\Delta G^\circ_f / RT) \quad \text{---(I)}$$

In the presence of catalyst -

$$k_f' = \frac{k_B T}{h} \exp(-\Delta G''_f / RT) \quad \text{---(II)}$$

From fig. it is evident that, the free energy of activation lowered for both forward and backward reactions without alternating the overall free energy change of the reaction. This means that a catalyst is able to change the rates of both the forward and backward reactions but ΔG° does not get changed.

Therefore the equilibrium constant will not get changed in the presence of the catalyst because ΔG° is related to k by the relation-

$$-\Delta G^\circ = RT \ln k$$

Mathematically,

Rate constant for the forward reaction -

$$k_f = \frac{k_B T}{h} \exp\left(\frac{-\Delta G'^\circ_f}{RT}\right) \quad \text{---(III)}$$

Rate constant for the backward reaction-

$$k_b = \frac{k_B T}{h} \exp\left(\frac{-\Delta G'^\circ_b}{RT}\right) \quad \text{---(IV)}$$

Now, (iii) ÷ (iv) we get—

$$K_{eq} = \frac{k_f}{k_b} = \frac{\frac{k_B T}{h} \exp\left(-\frac{\Delta G_f'^{\circ}}{RT}\right)}{\frac{k_B T}{h} \exp\left(-\frac{\Delta G_b'^{\circ}}{RT}\right)}$$

$$\Rightarrow K_{eq} = \exp\left[\left(-\frac{\Delta G_f'^{\circ}}{RT}\right) - \left(-\frac{\Delta G_b'^{\circ}}{RT}\right)\right]$$

$$\Rightarrow K_{eq} = \exp\left[-\Delta G^{\circ}/RT\right] \; ; \text{ where } \Delta G^{\circ} = \Delta G_f'^{\circ} - \Delta G_b'^{\circ}$$

Since, ΔG° has the same value in the presence or absence of a catalyst, K_{eq} remains the same.

Q. Discuss the kinetics of heterogeneous (gas-solid) unimolecular surface reaction. Write down its limiting cases.

Kinetics of heterogeneous unimolecular surface reaction:

If A is the reactant molecule and s is the surface atom of the solid, the elementary process may be depicted as follows—

$$A + S \underset{k_{-1}}{\overset{k_1}{\rightleftharpoons}} AS$$

$$AS \xrightarrow{k_2} \text{products}$$

Where AS refers to the adsorbed molecule. Suppose θ is the fraction of the surface covered by A at any time t and pressure p.

According to Langmuir–Hinshul wood, the reaction rate should be –

$$\text{Rate} = r = k_2\theta \quad \text{——(I)}$$

If we assume steady-state approximation for [AS] we get–

$$\frac{d[AS]}{dt} = k_1[A][S] - k_{-1}[AS] - k_2[AS] = 0$$

$$\Rightarrow [AS] = \frac{k_1[A][S]}{k_{-1} + k_2} \quad \text{——(II)}$$

Now, Let concentration of vacant sites $[S] = c_s(1-\theta)$

Concentration of occupied sites, $[AS] = c_s\theta$

Where, c_s refers to the total concentration of the surface of the catalyst.

Substituting the values of [S] and [AS] in equation (II) we get,

$$c_s\theta = \frac{k_1[A]c_s(1-\theta)}{k_{-1} + k_2}$$

$$\Rightarrow \theta = \frac{k_1[A]}{k_1[A] + k_{-1} + k_2} \quad \text{——(III)}$$

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putting the values of θ in equation (i) we get—

$$r = \frac{k_2 k_1 [A]}{k_1 [A] + k_{-1} + k_2}$$

$$\Rightarrow r = \frac{k_2 k_1 [A]}{k_1 [A]} + \frac{k_2 k_1 [A]}{k_1 + k_2}$$

$$\Rightarrow \frac{1}{r} = \frac{1}{k_2} + \frac{k_{-1} + k_2}{k_1 k_2 [A]} \quad \text{——— (iv)}$$

For gaseous reactions, [A] can be replaced by partial pressure and equation (iv) becomes—

$$\frac{1}{r} = \frac{1}{k_2} + \frac{k_{-1} + k_2}{k_2 k_1} \cdot \frac{1}{P_A} \quad \text{——— (v)}$$

A plot of $\frac{1}{r}$ against $\frac{1}{P_A}$ should be straight line having $\frac{1}{k_2}$ as the intercept and $\frac{k_{-1} + k_2}{k_2 k_1}$ as the slope.

Limiting cases:

Case – I:

In the rate expression from equation (v),

$$r = \frac{k_1 k_2 P_A}{k_1 P_A + k_{-1} + k_2}$$

When $k_2 \gg (k_1 P_A + k_{-1})$ then

$$r = k_1 P_A \quad \text{——— (vi)}$$

On integrating of (VI) we get –

$$k_1 = \frac{2.303}{t} \log_{10}\left(\frac{P_i}{P}\right)$$

Where P_i is the initial pressure and p its pressure at any time t. This equation is first order with respect to concentration of A.

Case II:

If $k_2 \ll (k_1 P_A + k_{-1})$ then we get –

$$r = \frac{k_1 k_2 P_A}{k_1 P_A + k_{-1}}$$

$$= \frac{(k_1/k_{-1}) k_2 P_A}{(k_1/k_{-1}) P_A + 1}$$

$$= \frac{k_a k_2 P_A}{k P_A + 1} \qquad \text{(VII)}$$

Where, k refers to adsorption equlibrium constant.

Equation (VII) is identical with Langmuir adsorption isotherm which is further analysed as follows –

a. At low pressure i.e $kP_A < 1$

$$r = k \cdot k_2 P_A$$

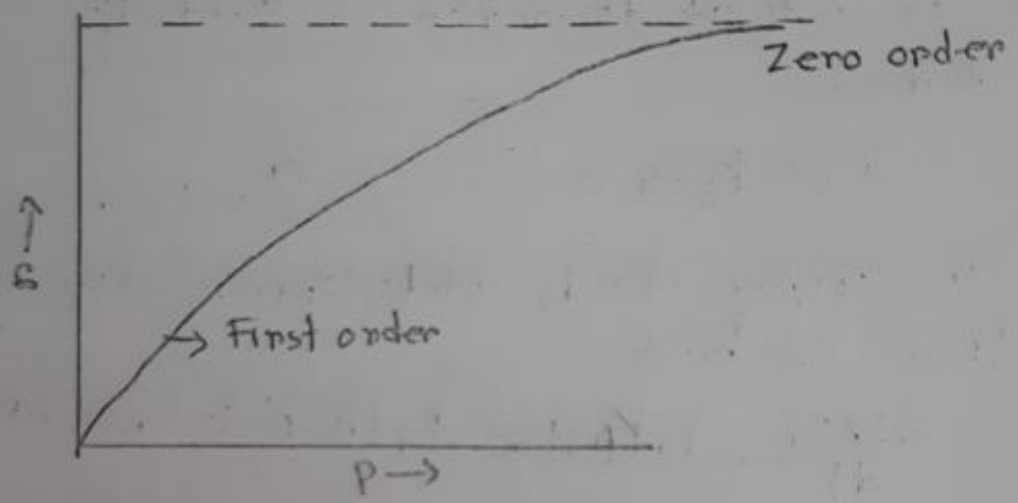
The reaction would be first order.

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b. At high pressure i.e $kP_A > 1$

$$r = k_2$$

Then the reaction would be zero order.



Q. Derive an expression for kinetics of bimolecular heterogeneous surface reactions. What happens if both reactants adsorbed very slightly and one reactant adsorbed very strongly? [2018] [2016]

Kinetics of bi-molecular surface reaction:

Consider a reaction in which two molecules A and B react on a surface and get absorbed on neighboring sites. The process may take places in two ways-

$$A + S \underset{k_{-1}}{\overset{k_1}{\rightleftharpoons}} AS$$

$$B + S \underset{k_{-1}}{\overset{k_1}{\rightleftharpoons}} BS$$

$$AS + BS \xrightarrow{k_3} \text{products}.$$

Suppose θ_1 and θ_2 be the fraction of the surface covered by adsorption of A and B respectively. The fraction of the surface which is vacant, $(1-\theta_1-\theta_2)$ The rate of formation of products is given as follows —

$$r = k_3 \theta_1 \theta_2 \quad \text{——— (1)}$$

On applying steady-state approximation to [AS] and [BS], we have—

$$\frac{d[AS]}{dt} = k_1[A][S] - k_{-1}[AS] - k_3[AS][BS] = 0 \quad \text{——— (2)}$$

$$\frac{d[BS]}{dt} = k_2[B][S] - k_{-2}[BS] - k_3[AS][BS] = 0 \quad \text{——— (3)}$$

Let, c_s is the total concentration of the surface sites of the catalyst. Therefore—

$$[AS] = c_s \theta_1$$
$$[BS] = c_s \theta_2$$
$$[S] = c_s(1-\theta_1-\theta_2)$$

On substituting the values [AS], [BS] and [S] in equation (2) and (3) we get—

$$k_1[A]c_s(1-\theta_1-\theta_2) = k_{-1}c_s\theta_1 - k_3 c_s^2 \theta_1\theta_2 = 0 \quad \text{——— (4)}$$

$$k_2[B]c_s(1-\theta_1-\theta_2) = k_{-2}c_s\theta_2 - k_3 c_s^2 \theta_1\theta_2 = 0 \quad \text{——— (5)}$$

If it is assumed that rate determining step is the chemical reaction between the adsorbed molecules, then $k_3 \to 0$ and equation (4) and (5) becomes as follows–

$$k_1[A](1-\theta_1-\theta_2) = k_{-1}\theta_1$$

$$k_2[B](1-\theta_1-\theta_2) = k_{-2}\theta_2$$

So, $$\theta_1 = \frac{k_1}{k_{-1}}[A](1-\theta_1-\theta_2)$$

$$= k_1[A][1-\theta_1-\theta_2] \quad \text{——— (6)}$$

Where $k_1 = k_1/k_{-1}$.

And, $$\theta_2 = \frac{k_2}{k_{-2}}[B](1-\theta_1-\theta_2)$$

$$= k_2[B](1-\theta_1-\theta_2). \quad \text{——— (7)}$$

Where $k_2 = k_2/k_{-2}$

On solving for θ_1 and θ_2 we obtain–

$$\theta_1 = k_1[A]/(1+k_1[A]+k_2[B])$$

$$\theta_2 = k_2[B]/(1+k_1[A]+k_2[B])$$

On inserting the value of θ_1 and θ_2 in equation (1) we get

$$r = k_3 k_1 k_2 [A][B]/(1+k_1[A]+k_2[B])^2 \quad \text{——— (8)}$$

When P_A and P_B denote the partial pressure of A and B for the gaseous reaction. Equation (8) becomes–

$$r = k_3 k_1 k_2 P_A P_B/(1+k_1 P_A + k_2 P_B)^2$$

Three special cases of important significance may arise from equation (9)

Case I:

If each gas (A and B) gets adsorbed very slightly, then $k_1 P_A \ll 1$ and $k_2 P_B \ll 1$, so

$$r = k_3 k_1 k_2 P_A P_B$$

This implies that the reaction will be second order.

Case II:

When one reactant, A, relatively more strongly ~~ab~~ adsorbed than B then, $k_1 P_A \gg k_2 P_B$, so

$$r = k_3 k_1 P_A k_2 P_B / (1 + k_1 P_A)^2$$

This implies that the reaction will be first order with respect to B.

Case III:

When one reactant A, very strongly adsorbed then, $k_1 P_A \gg k_2 P_B$ and $k_1 P_A \gg 1$, so

$$r = k_3 k_2 P_B / k_1 P_A$$

This implies that the rate will depend on the concentration of strongly adsorbed reactant.

Q. What is acid-base catalysis? What are the different types of acid-base catalysis? [2015]

Acid base catalysis: Acid base catalysis is a chemical reaction which is catalysed by acid or base and occurs by transfering a proton from an acid to substrate molecule or from substrate molecule to the base.

Types of acid base catalysis: Acid base catalysis are three types:

1. Specific hydrogen ion catalysis: These are the reaction which are catalysed by H^+ ion only.

For example, inversion of cane sugar —

$$C_{12}H_{22}O_{11} + H_2O \xrightarrow{H^+} C_6H_{12}O_6 + C_6H_{12}O_6$$

11. Specific hydroxyl ion catalysis: These are the reaction which are catalysez by OH^- ions only.

For example, conversion of acetone into diacetone

$$CH_3COCH_3 + CH_3COCH_3 \xrightarrow{OH^-} CH_3COCH_2C(CH_3)_2OH$$

(iii) Hydrogen and Hydroxyl ion catalysis: These are the reaction which are catalysed by both H^+ and OH^- ions simultaneously.

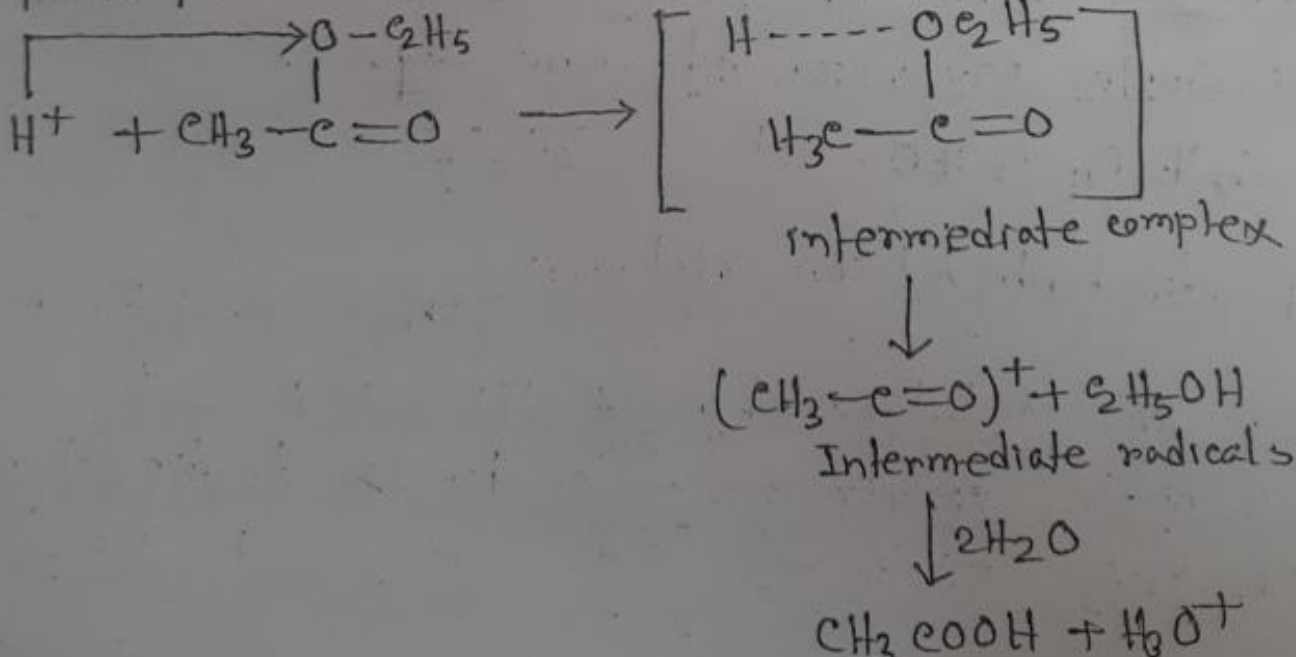
$$CH_3COOC_2H_5 + H_2O \xrightarrow{H^+ \text{ or } OH^-} CH_3COOH + C_2H_5OH$$

Q. Discuss the mechanism and kinetics of acid-base catalysis [2017] [2015]

Mechanism of acid base catalysis:

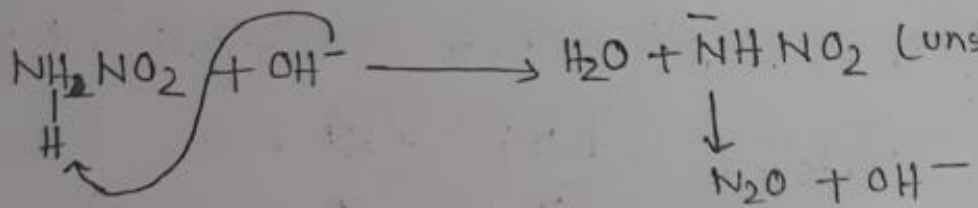
In acid catalysis, the H^+ forms an intermidiate complex with the reactant, which then reacts to give back the proton.

For example: Hydrolysis of Ethyl acetate catalysed by proton.



In base catalysis, the OH^- ion (or any Bronsted base) accepts a proton from the reactant to form an intermediate complex which then reacts or decomposes to regenerate the OH^-

For example, Decomposition of nitramide catalysed by OH^- ion



Kinetics of acid base catalysis: (2017)

Consider a acid base reaction -

$$S + HA \underset{k_{-1}}{\overset{k_1}{\rightleftharpoons}} SH^+ + A^- \quad \text{---------(I)}$$

Where, S is the substrate and SH^+ is the intermediate compound.

The intermediate compound then hydrolysed to give product

$$SH^+ + H_2O \xrightarrow{k_2} \text{product} \quad \text{---------(II)}$$

This reaction is slow and rate determining step

$\therefore$ rate of reaction $= k_2 [SH^+]$ ---------(III)

If we assume steady-state approximation for $[SH^+]$ we get –

$$\frac{d[SH^+]}{dt} = k_1[S][HA] - k_{-1}[SH^+][A^-] - k_2[SH^+] = 0$$

$$\Rightarrow [SH^+] = \frac{k_1[S][HA]}{k_{-1}[A^-] + k_2}$$

putting the values of $[SH^+]$ in equation (III) we get –

$$\text{Reaction rate, } r = \frac{k_1 k_2 [S][HA]}{k_2 + k_{-1}[A^-]} \qquad \text{(iv)}$$

Limitting cases:

(i) If $k_2 \gg k_{-1}[A^-]$, equation (iv) becomes –

$$r = \frac{k_1 k_2 [S][HA]}{k_2}$$

$$= k_1[S][HA]$$

(ii) If $k_2 \ll k_{-1} k_1 [A]$ then the equation (iv) becomes –

$$r = \frac{k_1 k_2 [S][HA]}{k_{-1}[A]}$$

This is the kinetics of a acid base reaction.

Q. Mention the effect of pH on rates of acid-base catalysed reaction with a suitable diagram.

Effect of pH on reaction rate of acid-base catalysed reaction:

We know that the rate constant of a reaction catalysed by H^+ is proportional to $[H^+]$. i.e

$$k = k_{H^+}[H^+] \quad \text{---------- (I)}$$

Where, k_{H^+} is the catalytic co-efficient of the hydrogen ions

Taking logarithm of equation (I)

$$\log k = \log k_{H^+} + \log [H^+]$$

$$\Rightarrow \log k = \log k_{H^+} - pH \quad \text{---------- (II)} \quad [\because pH = -\log [H^+]]$$

From equation (II) it is seen that a plot of $\log k$ vs pH should be straight line having slope equal to -1 which is shown in fig $-1(a)$

For a reaction catalyzed by OH^- ions.

$$k = k_{OH^-}[OH]$$

$$= k_{OH^-} \frac{k_w}{[H^+]} \quad \text{---------- (III)}$$

Where, k_{OH^-} is the catalytic co-efficient of OH^-

Taking logarithm of equation (III)

$$\log k = \log k_{OH^-} + \log k_w - \log [H^+],$$

$$\Rightarrow \log k = \log k_{OH^-} + \log k_w + pH \quad \text{——— (iv)}$$

Hence a plot of $\log k$ against pH should be straight line having a slope equal to +1. which is shown in fig-1(b)

For a reaction catalysed by both acid and base—

$$k = k_0 + k_{H^+}[H^+] + k_{OH^-}[OH^-] \quad \text{——— (v)}$$

In this case, $\log k$ vs pH show minima value which is given by—

$$(pH)_{min} = \frac{1}{2}[\log k_{H^+} - \log_{OH^-} + pk_w]$$

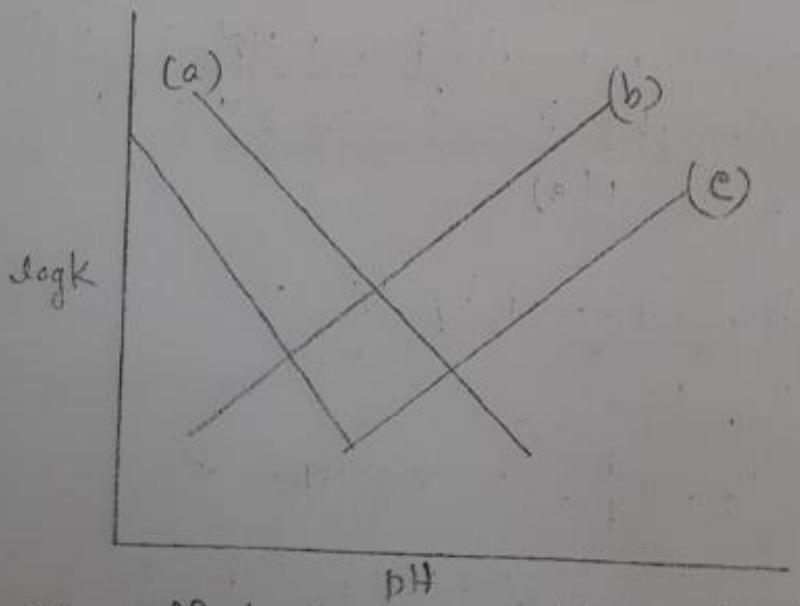


Fig.: The effect of pH on acid base reaction.

a. Specific H^+ ion catalysed.

b. Specific OH^- ion catalysed.

c. Both H^+ and OH^- catalysed acid-base reactions.

Q. Describe absolute rate theory in heterogeneous gas reaction.

Let us consider a bimolecular reaction between A and B involving a reaction centres. Then—

$$A + B + S \rightleftharpoons [A-B-S] \neq \text{products} \quad \text{——— (I)}$$

In this type of reactions there exists an equilibrium between the reactants and the activated complex. If c_A, c_B, c_S denote the concentration of A, B and S respectively, then applying the law of mass action—

$$K^{\neq} = \frac{c^{\neq}}{c_A c_B c_S} \quad \text{——— (II)}$$

If we consider the partition function equation (II) becomes

$$K^{\neq} = \frac{\theta^{\neq}}{\theta_A \theta_B \theta_S} e^{-\varepsilon/RT} \quad \text{——— (III)}$$

Comparing equation (II) and (III) we get—

$$\frac{c^{\neq}}{c_A c_B c_S} = \frac{\theta^{\neq}}{\theta_A \theta_B \theta_S} e^{-E/RT}$$

$$\Rightarrow c^{\neq} = c_A c_B c_S \cdot \frac{\theta^{\neq}}{\theta_A \theta_B \theta_S} e^{-\varepsilon/RT} \quad \text{——— (IV)}$$

Now, rate of a bimolecular reaction—

$$\frac{dx}{dt} = c^{\neq} \frac{kT}{h}$$

$$\Rightarrow \frac{dx}{dt} = c_A c_B c_S \cdot \frac{\theta^{\neq}}{\theta_A \theta_B \theta_S} \cdot e^{-\varepsilon/RT} \frac{kT}{h} \quad \text{——— (V)}$$

Again, reaction rate may also be written as –

$$\frac{dx}{dt} = k_r \cdot C_A C_B \quad \text{——(vi)}$$

From (v) and (vi) we get –

$$k_r \cdot C_A C_B = C_A C_B C_s \cdot \frac{\theta^{\neq}}{\theta_A \theta_B \theta_s} \cdot e^{-\varepsilon/RT} \cdot \frac{kT}{h}$$

$$\Rightarrow k_r = \frac{kT}{h} C_s \frac{\theta^{\neq}}{\theta_A \theta_B \theta_s} \cdot e^{-\varepsilon/RT}$$

Since s part of the surface possess no translational rotational contribution. $\frac{\theta^{\neq}}{\theta_s}$ may be taken as unity

$$k_r = \frac{kT}{h} \cdot C_s \frac{1}{\theta_A \theta_B} \cdot e^{-\varepsilon/RT}$$

Q. Describe the effect of temperature on heterogeneous reactions:- [2017]

Arrhenius equation for heterogeneous reaction is—

$$\frac{d \ln k}{dT} = \frac{E_a}{RT^2}$$

For unimolecular surface reaction—

1. If the reactant is strongly adsorbed—

$$\text{rate} = r_1 = k_2$$

In this case $E_a = E_f$

II. If the reactant is slightly adsorbed —

$$\text{rate} = r = k \cdot k_2 P_A$$

From Arrhenius equation—

$$\frac{d \ln (k \cdot k_2)}{dT} = \frac{E_a}{RT^2}$$

$$\Rightarrow \frac{d \ln K}{dT} + \frac{d \ln k_2}{dT} = \frac{E_a}{RT^2}$$

$$\Rightarrow \frac{\Delta H_{ads}}{dT} + \frac{E_f}{dT} = \frac{E_a}{RT^2}$$

As, ΔH_{ads} is a negative quantity, then $E_f > E_a$ i.e activation energy of this type reaction get lowered.

For bimolecular reaction —

(i) If the reactant is slightly adsorbed, then

$$r = k_3 \cdot K_1 K_2 P_A P_B$$

From Arrhenius equation—

$$\frac{d \ln (k_3 \cdot K_1 \cdot K_2)}{dT} = \frac{E_a}{RT^2}$$

$$\Rightarrow \frac{d \ln k_3}{dT} + \frac{d \ln K_1}{dT} + \frac{d \ln K_2}{dT} = \frac{E_a}{RT^2}$$

$$\Rightarrow \frac{E_f}{dT} + \frac{(\Delta H_{ads})_1}{dT} + \frac{(\Delta H_{ads})_2}{dT} = \frac{E_a}{RT^2}$$

So, $E_f > E_a$, as ΔH_{ads} is negative quantity.

(ii) If the reactant is strongly adsorbed, then

$$r = k_3 \cdot \frac{K_2}{K_1} \frac{P_B}{P_A}$$

From Arrhenius equation—

$$\frac{d \ln \left(k_3 \cdot \frac{K_2}{K_1}\right)}{dT} = \frac{E_a}{RT^2}$$

$$\Rightarrow \frac{d \ln k_3}{dT} + \frac{d \ln K_2}{dT} - \frac{d \ln K_1}{dT} = \frac{E_a}{RT^2}$$

$$\Rightarrow \frac{E_f}{dT} + \frac{(\Delta H_{ads})_2}{dT} - \frac{(\Delta H_{ads})_1}{dT} = \frac{E_a}{RT^2}$$

It mean that strongly adsorbed product increase the activation energy.

SURFACE CHEMISTRY

Q. Explain the terms : Adsorption, absorption and Sorption with diagram.

Adsorption:

Adsorption is a surface phenomenon. The surface of a solid has a tendency to attract and to retain molecules of other species with which such surfaces come in contact. This phenomenon of surfaces termed as adsorption.

Example: Adsorption of ammonia by charcoal.

Absorption:

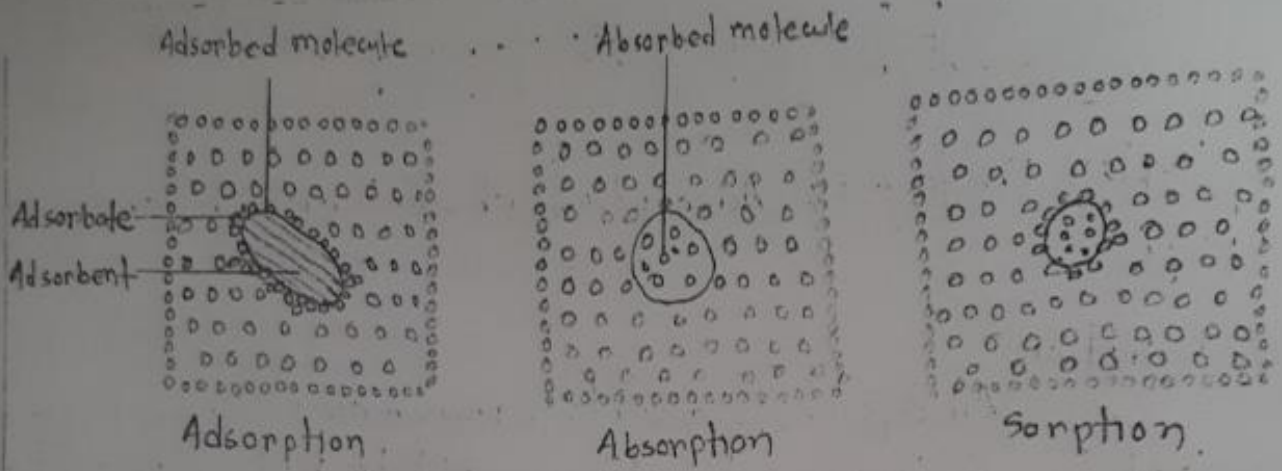
Absorption is a bulk phenomenon in which the substance assimilated is uniformly distributed throughout the body of a solid or liquid to form a solution or compound.

Example: Absorption of water vapour by anhydrous calcium chloride.

Sorption:

The process in which both adsorption and absorption take place simultaneously is termed as sorpti

Example : The charcoal adsorbed as well as absorb hydrogen gas.



Q. What do you mean by Adsorbate and Adsorbent?

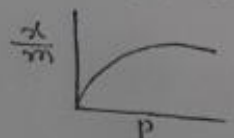
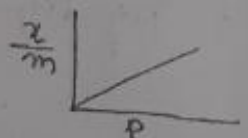
Adsorbate:
The substance which gets adsorbed is called adsorbate.

Example: There are various gases (H_2, N_2, O_2, Ne, SO_2, NH_3 etc).

Adsorbent: The substance which absorbs is called adsorbent.

Example: Silica gel, charcoal, metals (Ni, Cu, Ag, Pd, Pt etc).

Q. Differentiate between adsorption and absorption.

Adsorption	Absorption
I. Accumulation of the molecular species at surface not in the bulk.	I. It is the process by which atoms, molecules or ions enters the bulk.
II. Adsorption is a surface phenomenon.	II. Absorption is a bulk phenomenon.
III. It is exothermic process	III. It is endothermic process
IV. The equilibrium is easily attained in a very short time	IV. The equilibrium takes place slowly.
V. It is a temperature dependent phenomenon.	V. It is not affected by temperature.
VI. For adsorption, $\frac{x}{m}$ vs p curve should be exponential. 	VI. For absorption, x/m vs p curve should be straight line 
VII. Example: Adsorption of ammonia by charcoal.	VII. Example: Absorption of water vapour by anhydrous calcium chloride

Q. Describe the ~~charisties~~ Characteristics of adsorption.

~~Charisties~~
Characteristics of adsorption:

I. Adsorption refers to the existance of a higher concentration of any particular component at the surface of liquid or a solid phase.

II. Adsorption is accompanied by decrease in the ΔG (free energy change) of the system when $\Delta G = 0$ adsorption equilibrium is said to be established.

III. Adsorption is invariably accompanied by evolution of heat i.e It is an exothermic process. In other words ΔH of adsorption is always negative.

IV. When a gas is adsorbed, the freedom of movement of its molecules becomes restricted. On account of it decrease in the entropy of the gas after adsorption, i.e ΔS is negative.

Q. Describe the factors which affecting adsorption.

The factors affecting adsorption:

I. Adsorbent:

Every solid surface has the adsorption capacity

II. Adsorbate:

a. Increasing solubility decreases adsorption.

b. Solubility of adsorbate is affected by molecular size, ionization and polarity.

III. pH:

- Affects surface change of the adsorbent and the change on the solute.

- Lowering pH increase adsorption for the organic material.

IV. Temperature:

As adsorption is an exothermic process, hence increasing temperature decreases adsorption.

V. pressure:

pressure leads to increase of adsorption and decrease of pressure causes desorption.

Q. Describe the classification of adsorption.
or, Define the terms - physisorption and chemisorption.

There are two types of adsorption -

1. physisorption,
11. chemisorption.

1. physisorption:

This is due to the gas molecules being held to the solid surface by van der Waal's attractive forces. It is also referred to as van der Waal's adsorption.

For example; adsorption of hydrogen or oxygen on charcoal is physisorption.

11. Chemisorption:

In this kind of adsorption, the gas molecules or atoms are held to the solid surface by chemical bonds. These bonds ~~are~~ may be covalent or ionic in nature.

For example; hydrogen is chemisorbed by nickel. Hydrogen molecule is first adsorbed by van der waals forces and then dissociates. The hydrogen atoms are thus chemisorbed on nickel.

Q. Differentiate between physisorption and chemisorption.

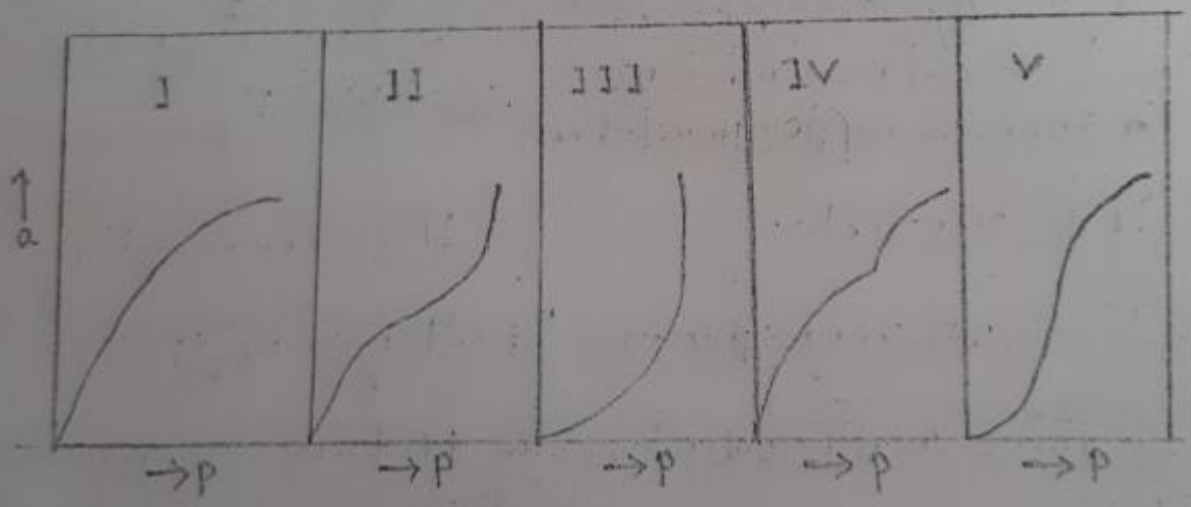
physisorption	Chemisorption
1. Low heat of adsorption usually in the range of 20–40 kJ mol^{-1}	1. High heat of adsorption in the range of 40-400 kJ mol^{-1}
2. Force of attraction are van der Waals force.	2. Forces of attraction are chemical bond forces.
3. It usually takes place at low temperature and decreases with increasing temperature.	3. It takes place at high temperature.
4. It is reversible	4. It is irreversible
5. It is not very specific	5. It is very specific
6. It forms multi molecular layers.	6. It forms mono molecular layers.
7. It does not requir any activation energy.	7. It requires activation energy.

Q. What is adsorption isotherm? show different types of adsorption isotherm with the help of a diagram.

Adsorption isotherm:

If the temperature is maintained constant and pressure is changed, then the curve between the amount of gas adsorbed and pressure is known as adsorption isotherm.

Types of adsorption Isotherms:



Type I:

This type of curve is obtained where mono molecular layer is formed on the surface of the adsorbent.

Example: Adsorption of nitrogen on charcoal at $-195^{\circ}C$

Type II:

This type of isotherm has a transition point 'B' which represents the pressure at which the formation of monomolecular layer is complete and that of the multimolecular layer is being started.

Example: Adsorption of nitrogen on silica gel at −195°C.

Type III:

In this isotherm, there is no transition point. Here the multimolecular formation starts even before the formation of monomolecular layer is complete.

Example: Adsorption of bromine or iodine vapours on silica gel at 79°C.

Type IV:

In this case, there is a tendency for a saturation state to be reached in the multimolecular region as well.

Example: Adsorption of benzene vapours on ferric oxide gel at 50°C.

Type - V:

This isotherm indicates multimolecular layer formation in the beginning.

Example: Adsorption of water vapour on charcoal at 100°C.

a. Describe Freundlich isotherm with suitable curve and give some limitations of this isotherm.

Freundlich proposed an empirical relation in the form of a mathematical equation—

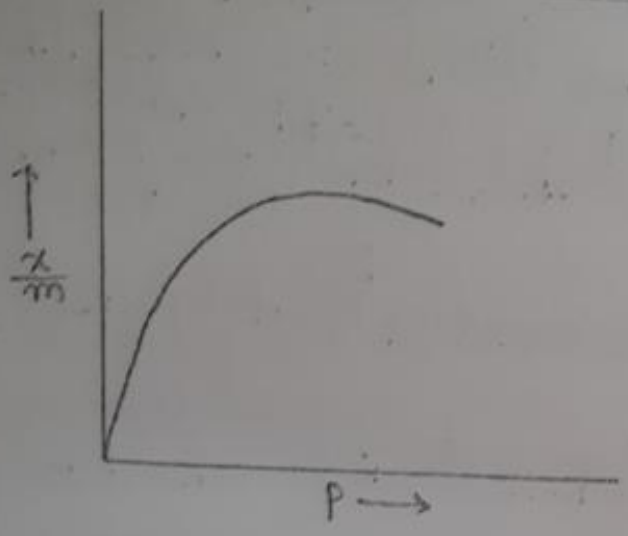
$$\frac{x}{m} = kp^{1/n} \quad \text{——— (I)}$$

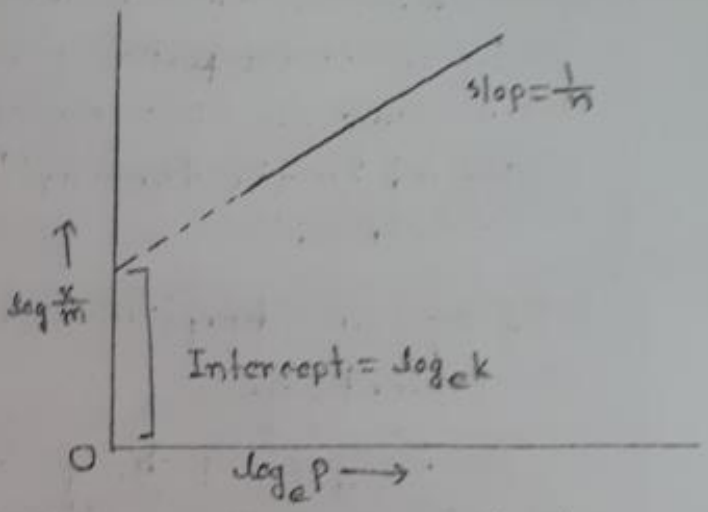
Where x is the mass of the gas adsorbed on a mass m of adsorbent at a pressure p. k and n are constant depending on the nature of the gas, the adsorbent and on temperature.

This relation is generally represented in the form of a curve obtained by plotting the mass of the gas adsorbed per unit mass of adsorbent (x/m) against equilibrium pressure.

Freundlich isotherm is not applicable at high pressure.
Taking logarithm of on both sides of equation (1) we get—

$$\log \frac{x}{m} = \frac{1}{n} \log p + \log k \quad \text{——— (II)}$$





→ This equation is for a straight line Thus a plot of log x/m against log p should be a straight line with slope 1/n and intercept log k.

→ However, it is actually found that the plots were straight lines at low pressures, while at high pressure they showed a slight curvature, especially at low temperature

→ This indicates that Freundlich' equation is approximate and does not apply to adsorption of gases by solids at higher pressures.

Limitations:

I. It is empirical formula without theoretical foundation.

II. It is valid over a certain range of pressure.

III. The constant k and n vary with temperature

Q. What are the assumptions of Langmuir adsorption isotherm? Derive the equation for Langmuir adsorption isotherm for the amount of a gas adsorbed on a solid surface. What would be the form of this isotherm when the gas pressure is varied?

Langmuir adsorption isotherm is based on the following assumptions—

i. The layer of the gas adsorbed on the solid adsorbent in one molecule thick.

ii. The adsorbed layer is uniform all over the adsorbent.

iii. The adsorption is taking place on the fixed sites and there is no interaction between the addacent adsorbed molecules.

iv. The process of adsorption is a dynamic process which consists of two oppsing processes.

   a. condensation process.

   b. Evaporation process.

a. Condensation process: It involves the condensation of gas molecules on the surface of the adsorbent.

b. Evaporation process:

→ It involves the evaporation of gas molecules from the surface of the adsorbent.

→ When the equilibrium is reached, the rate of condensation becomes equal to the rate of evaporation.

Derivation:

Let, n is the number of molecules of the gas striking and α is the fraction of the molecules condensing on the surface. Then, αn is the total number of the molecules which condense on the surface.

∴ Rate of condensation of gaseous molecules on the surface $= (1-\theta)\alpha n$

Where, θ is the fraction of the area covered by the gas molecules.

We know that:

Rate of evaporation $\propto$ area of surface covered

$\Rightarrow$ Rate of evaporation $= k\theta$

Where k is proportionality constant.

At equilibrium, the rate of condensation is equal to the rate of evaporation—

Thus—

$$(1-\theta)\alpha n = k\theta$$

$$\Rightarrow \alpha n - \theta\alpha n = k\theta$$

$$\Rightarrow \alpha n = k\theta + \alpha\theta n$$

$$\Rightarrow \theta = \frac{\alpha n}{k+\alpha n} \quad \text{——— (1)}$$

Again, number of molecules $\propto$ pressure of the gas striking the surface

$\Rightarrow n \propto p$

$\Rightarrow n = \beta p$

Where, β is proportionality constant.

Now, $$\frac{k+\alpha n}{\alpha n} = \frac{k}{\alpha n} + 1$$

$$= \frac{k}{\alpha \beta p} + 1 \quad [\because n = \beta p]$$

$$= \frac{1}{k_1 p} + 1 \quad \left[\frac{\alpha\beta}{k} = k_1\right]$$

$$\Rightarrow \frac{k+\alpha n}{\alpha n} = \frac{1+k_1 p}{k_1 p}$$

$$\Rightarrow \frac{\alpha n}{k+\alpha n} = \frac{k_1 p}{k_1 p + 1} \quad \text{——— (II)}$$

From equation (I) and (II) we get—

$$\theta = \frac{k_1 p}{1+k_1 p}$$

If the adsorbed layer is assumed to be unimolecular in thickness—

$$\frac{x}{m} \propto \theta$$

$$\Rightarrow \frac{x}{m} = k_2 \theta$$

$$\Rightarrow \frac{x}{m} = \frac{k_1 k_2 p}{1+k_1 p}$$

$$\Rightarrow \frac{1}{x/m} = \frac{1+k_1 p}{k_1 k_2 p} \Rightarrow \frac{\theta_1}{x/m} = \frac{1}{k_1 k_2 p} + \frac{k_1 p}{k_1 k_2 p} \Rightarrow \frac{p}{x/m} = \frac{p}{k_1 k_2 p} + \frac{k_1 p}{k_1}$$

$$\Rightarrow \frac{p}{x/m} = \frac{1}{k_1 k_2} + \frac{p}{k_2} \quad \text{——— (III)}$$

Equation (III) is known as Langmuir adsorption isotherm.

Forms of Langmuir Isotherm when the gas pressure is varied :

From langmuir isotherm —

$$\frac{x}{m} = \frac{k_1 k_2 P}{1 + k_1 P}$$

$$\Rightarrow \frac{x}{m} = \frac{aP}{1+bP} \quad \left[\text{let, } \begin{matrix} k_1 k_2 = a \\ k_1 = b \end{matrix}\right]$$

At high pressure —

$$\frac{x}{m} = \frac{aP}{1+bP}$$

$$\Rightarrow \frac{m}{x} = \frac{1+bP}{aP}$$

$$= \frac{1}{aP} + \frac{b}{a}$$

If the pressure is very high then the value of $\frac{1}{aP}$ decreases and almost zero

$$\frac{1}{aP} \approx 0$$

then, $\frac{m}{x} = \frac{b}{a}$

$$\Rightarrow \frac{x}{m} = \frac{a}{b}$$

This is the form of langmuir adsorption isotherm at high pressure.

At low pressure :

$$\frac{x}{m} = \frac{aP}{1+bP}$$

When the pressure is very low then $1+bP \simeq 1$ and the value of $\frac{x}{m}$ is directly proportional to P

$$\frac{x}{m} \propto P$$

Between the high and low and high values of P, the above equation becomes —

$$\frac{x}{m} = kp^{1/n}$$

which is the Freundlich equation.

Q. Describe the limitations of Langmuir adsorption isotherm.

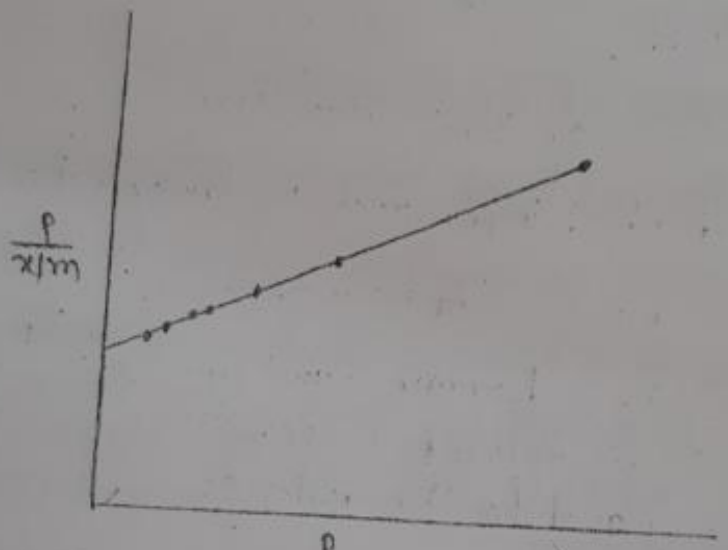
Limitations:

I. Langmuir postulated that a saturated value of adsorption is independent on the temperature. But experiment shows that it is actually falling off with rising temperature.

II. Langmuir assumed that an adsorption film on a plane surface will never be over a molecule thick. In actual practice, much thicker films have been reported.

III. This theory can not explain the five types of adsorption isotherms.

* A plot of $p/x/m$ against p should be a straight line



[must add after derivation of L.A.I.T].

Q. What is the basic principle of B.E.T equation and explain the meaning of various terms involved in the equation.

Basic principles of B.E.T equation:

I. The adsorbed layer may be polymolecular in thickness.

II. Langmuir's assumption applies to every layer.

III. There is a dynamic equilibrium between successive layers. The rate of evaporation from the first layer is equal to the rate of condensation on the preceding layer.

IV. The heat of adsorption in each layer is involved in each of the evaporation process.

V. After the first layer, the heat of adsorption is equal to latent heat of condensation of vapour.

VI. The derivation is based on the same kinetic picture as represented by Langmuir.

VII. Condensation forces are the principal forces in adsorption.

The form of B.E.T equation –

$$\frac{x}{(1-x)v} = \frac{1}{cV_m} + \frac{(c-1)x}{cV_m}$$

Where,

$$x = \frac{P}{P_0}$$

Here,

P = equilibrium pressure of the gas over the surface

P_0 = Saturated vapour pressure of the gas at experimental temperature.

x = Amount of gas adsorbed

V_m = Volume of gas adsorbed when the entire surface of the adsorbent is covered by a unimolecular layer.

Q. Derive B.E.T equation for adsorption isotherm.

Derivation of B.E.T equation for adsorption isotherm:

Let, $S_0, S_1, S_2, \ldots S_i$ represent the surface area of adsorbent covered by 0, 1, 2, ... i layers of the adsorbed molecules. As S_0 must remain constant at equilibrium, the rate of evaporation from the first layer is equal to the rate of condensation on the preceding layer.

Thus—

$$a_1 P S_0 = b_1 S_1 e^{-E_1/RT} \quad \text{——— (i)}$$

For the equilibrium in successive layers, we have—

$$a_2 P S_1 = b_2 S_2 e^{-E_2/RT} \quad \text{——— (ii)}$$

$$a_3 P S_2 = b_3 S_3 e^{-E_3/RT} \quad \text{——— (iii)}$$

— — — — — — — — — —

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$$a_i P S_{i-1} = b_i S_i e^{-E_i/RT} \quad \text{——— (iv)}$$

Where, $a_1, a_2, \ldots a_i$ and $b_1, b_2, \ldots b_i$ are constants. $E_1, E_2, E_3 \ldots E_i$ represents the heat of adsorption of 1, 2, 3 ... i layers.

Now, total surface area A of the adsorbent can be written as—

$$A = \sum_{i=0}^{\infty} S_i$$

And, the volume adsorbed V is given by—

$$V = V_0 \sum_{i=0}^{\infty} i S_i$$

$$\therefore \frac{V}{A} = \frac{V_0 \sum_{i=0}^{\infty} i S_i}{\sum_{i=0}^{\infty} S_i}$$

$$\Rightarrow \frac{V}{AV_0} = \frac{\sum_{i=0}^{\infty} i S_i}{\sum_{i=0}^{\infty} S_i}$$

$$\Rightarrow \frac{V}{V_m} = \frac{\sum_{i=0}^{\infty} i S_i}{\sum_{i=0}^{\infty} S_i} \quad \text{——— (v)} \quad [V_m = AV_0]$$

From B.E.T assumptions, we know—

$$E_2 = E_3 = E_4 \dots E_i = E_L \quad \text{——— (VI)}$$

And, $$\frac{b_2}{a_2} = \frac{b_3}{a_3} \dots \frac{b_i}{a_1} = g \quad \text{——— (VII)}$$

Now, from equation (1) we get—

$$a_1 p S_0 = b_1 S_1 e^{-E_1/RT}$$

$$\Rightarrow S_1 = \frac{a_1}{b_1} \frac{p}{e^{-E_1/RT}} S_0$$

$$\Rightarrow S_1 = \left(\frac{a_1}{b_1}\right) p e^{E_1/RT} S_0$$

$\Rightarrow S_1 = y S_0$ ——— (VIII) ; where $y = \frac{a_1}{b_1} p e^{E_1/RT}$

From equation (II) we get,

$a_2 p S_1 = b_2 S_2 e^{-E_2/RT}$

$\Rightarrow S_2 = \frac{a_2}{b_2} \frac{p}{e^{-E_2/RT}} S_1$

$\Rightarrow S_2 = \frac{a_2}{b_2} p e^{E_2/RT} S_1$

$\Rightarrow S_2 = \left(\frac{1}{b_2/a_2}\right) p e^{E_2/RT} S_1$ $[\because E_2 = E_1]$

$\Rightarrow S_2 = \frac{p}{q} e^{E_1/RT} S_1$ $\left[\because \frac{b_2}{a_2} = q\right]$

$\Rightarrow S_2 = x S_1$ $\left[\text{where, } x = \frac{p}{q} e^{E_1/RT}\right]$

$\Rightarrow S_2 = xy S_0$ ——— (IX) [Using equation (VIII)]

Similarly,

$S_3 = x^2 y S_0$

$S_4 = x^3 y S_0$

- - - - - - - - -

$S_i = x^{i-1} y S_0$

$= \frac{x^i}{x} y S_0$

$= x^i \left(\frac{y}{x}\right) S_0$

$= x^i c S_0$ ——— (X) $\left[\because \frac{y}{x} = c\right]$

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Now, equation (v) becomes-

$$\frac{V}{V_m} = \frac{\sum_{i=0}^{\infty} i s_i}{\sum_{i=0}^{\infty} s_i}$$

$$\therefore \frac{\sum_{i=0}^{\infty} i c x^i s_0}{\sum_{i=0}^{\infty} c x^i s_0}$$

$$\Rightarrow \frac{V}{V_m} = \frac{c s_0 \sum_{i=1}^{\infty} i x^i}{s_0 \left[1 + c \sum_{i=0}^{\infty} x^i\right]}$$

$$\Rightarrow \frac{V}{V_m} = \frac{c \cdot \frac{x}{(1-x)^2}}{\left[1 + c \frac{x}{1-x}\right]} \quad \left[\because \sum_{i=1}^{\infty} i x^i = \frac{x}{(1-x)^2}, \ \sum_{i=0}^{\infty} x^i = \frac{x}{(1-x)}\right]$$

$$\Rightarrow \frac{V}{V_m} = \frac{cx}{(1-x)[1+(c-1)x]}$$

After rearranging, we get -

$$\frac{x}{(1-x)V} = \frac{1}{cV_m} + \frac{(c-1)x}{cV_m}$$

This equation is known as B.E.T equation for adsorption isotherm.

Q. Explain the terms - (i) Surface excess (ii) surface free energy.

Surface excess:

Surface excess is the difference between the amount of a component actually present in the system and the amount that would be present at the same temperature and pressure in the absence of adsorption.

If α and β are the two phases are in contact and the system contains several components J, each of which has an amount n_J.

If one component accumulate at the interface, the total amount J differs from the sum of $n_J(\alpha)$ and $n_J(\beta)$ by —

$$n_J(\sigma) = n_J - \{n_J(\alpha) + n_J(\beta)\}$$

This difference is expressed in terms of surface excess Γ_J —

$$\Gamma_J = \frac{n_J(\sigma)}{\sigma} \quad \text{——(1)}$$

From equation (1), it is said that, surface excess can be defined as an amount per unit of area of the surface.

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Surface free energy:

Surface free energy (SFE) can be considered as the surface tension of solid. Surface free energy of the material arises from molecular interaction at the solid surface.

For example, in metals, the atoms are linked together with extremly strong metal bondss. The surface free energy of metal is thus typically very high.

Then, on the other hand, bonds in polymeric materials are typically weak which makes the SFE of polymers low.

Surface free energy determines how the solids behaves in contact with other materials. From a thermodynamics point of view, the high surface free energy materials are not stable and there is a need to minimize it.

Q. What is meant by surface tension of a liquid? Explain why liquids rise up in capillary tube when the tube is placed in a liquid. Derive an equation for the surface tension of a liquid.

Surface tension:

Surface tension is the energy required to change the surface area of a liquid by a unit amount of length.

Thermodynamically, surface tension γ is defined as work done to change the surface area, σ.

Thus, surface tension, $\gamma = \frac{dw}{d\sigma}$ ——— (1)

Where, $d\sigma$ is the change in surface area and dw is the work done for changing area.

But, at constant volume and temperature, the work done is identical with change in Gibbs energy and Helmholtz energy—

So, surface tension, $\gamma = \frac{dw}{d\sigma} = \frac{dG}{d\sigma} = \frac{dA}{d\sigma}$

Reason behind liquid rise up in capillary tube when the tube is placed in a liquid:

If a capillary tube is placed vertically into a liquid, capillary action will rise or suppress the liquid inside the tube depending on the combination of substances. The actual effect depends on the relative strength of the cohesive and adhesive forces and the contact angle θ. If θ is less than 90° then the fluid will be rised, if θ is greater than 90° then it will be suppressed.

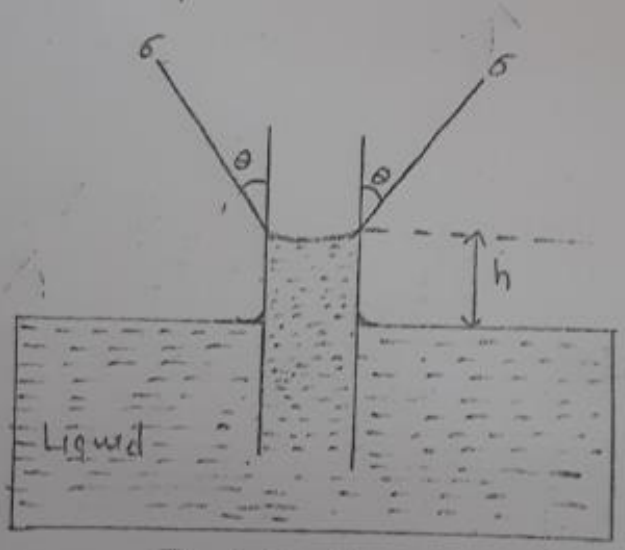


Fig.: capillary rise

In the case of placed a capillary tube in water then rise of liquid surface happen which is called capillary rise. Here Adhesion force is greater than ~~adhesion~~ cohesion force and the contact angle θ, which is defined as the angle between the surface tension and vertical is less than 90°. For this two reason the liquid wet the glass surface and the liquid level will arise.